Problem N. Rotting Oranges

OS Linux

You are given a 2D matrix consisting of oranges.
mat[i][j] = 0: Empty cell
mat[i][j] = 1: Fresh Orange
mat[i][j] = 2: Rotten Orange

As a matter of fact, the rotten oranges spread to the neighbouring cells and rot the fresh oranges. Assuming it takes 1 day for a rotten orange to spread to a neighbouring fresh orange, find out the minimum number of days in which all the oranges rot out. If there is atleast 1 fresh orange which will never rot, print -1.
2 cells are considered neighbours if they share a common side.

Input Format

First line of input contains T - number of test cases. First line of each test case contains N - the size of the matrix (NxN). Its followed by NxN denoting the state of the matrix.

Constraints

1 <= T <= 100
1 <= N <= 500
0 <= mat[i][j] <= 2

Output Format

For each test case, print the minimum number of days in which all the oranges rot out, separated by new line.

Input	Output
2 3 021 210 111 2 01 20	3 -1

Explanation 0

Self Explanatory